

AP Police SI Prelims 2026 — Deep Dive:

Complete Arithmetic

A full walkthrough of every Arithmetic topic in the syllabus: concept, formulas, shortcuts, and solved problems. Work through each example with pen and paper before reading the solution.

1. Number System

Core: Natural, whole, integers, rational/irrational, prime, composite. **Divisibility:** 2 (last digit even), 3 & 9 (digit sum), 4 (last two digits), 5 (ends 0/5), 11 (alternating-digit-sum), 6 (by 2 and 3). **LCM & HCF:** $\text{LCM} \times \text{HCF} = \text{product of two numbers}$. **Sum formulas:** $1+2+\dots+n = n(n+1)/2$; squares = $n(n+1)(2n+1)/6$; cubes = $[n(n+1)/2]^2$.

Solved 1: Find the HCF and LCM of 12 and 18. $\rightarrow 12 = 2^2 \times 3$, $18 = 2 \times 3^2$. $\text{HCF} = 2 \times 3 = 6$; $\text{LCM} = 2^2 \times 3^2 = 36$. Check: $6 \times 36 = 216 = 12 \times 18$. ✓

Solved 2: Smallest number divisible by both 4 and 6. $\rightarrow \text{LCM}(4,6) = 12$.

2. Percentage

Core: $x\% = x/100$. To increase by $x\%$, multiply by $(1+x/100)$. **Key results:** Rise $x\%$ then fall $x\% \rightarrow \text{net loss } (x^2/100)\%$. To keep spending constant when price rises $x\%$, cut use by $x/(100+x) \times 100\%$.

Solved 1: A's salary rises 10% then 20%. Net rise? $\rightarrow 1.10 \times 1.20 = 1.32 \rightarrow 32\%$ rise.

Solved 2: If 40% of a number is 120, find the number. $\rightarrow N = 120/0.40 = 300$.

Solved 3: Price of sugar rises 25%. To keep the bill the same, by what % should consumption drop? $\rightarrow 25/(125) \times 100 = 20\%$.

3. Ratio & Proportion

Core: $a:b = c:d \Rightarrow ad = bc$. Combine $a:b$ and $b:c$ by making b common. **Splitting:** Divide A in $m:n \rightarrow A \cdot m/(m+n)$ and $A \cdot n/(m+n)$.

Solved 1: Divide ₹800 between A and B in ratio 3:5. $\rightarrow A = 800 \times 3/8 = 300$; $B = 800 \times 5/8 = 500$.

Solved 2: $A:B = 2:3$, $B:C = 4:5$. Find $A:B:C$. $\rightarrow B$ common: $A:B = 8:12$, $B:C = 12:15 \Rightarrow 8:12:15$.

4. Average

Core: Average = sum/count. Average speed for equal distances at $a, b = 2ab/(a+b)$.

Solved 1: Average of 25, 30, 35, 40, 45? $\rightarrow \text{Sum } 175 / 5 = 35$.

Solved 2: Average of 10 numbers is 20; if each is increased by 5, new average? $\rightarrow 25$ (adding a constant shifts the average by it).

Solved 3: A travels 60 km at 30 km/h and back at 60 km/h. Average speed? $\rightarrow 2(30)(60)/(30+60) = 3600/90 = \mathbf{40 \text{ km/h}}$.

5. Profit & Loss

Core: Profit% = $(SP-CP)/CP \times 100$; Loss% = $(CP-SP)/CP \times 100$. Successive discounts x%, y% $\rightarrow x+y-xy/100$.

Solved 1: CP = 500, SP = 650. Profit%? $\rightarrow 150/500 \times 100 = \mathbf{30\%}$.

Solved 2: An article costing 800 is marked up 50% and sold at 20% discount. Profit%? $\rightarrow MP = 1200$; SP = $1200 \times 0.8 = 960$; profit = $160/800 \times 100 = \mathbf{20\%}$.

Solved 3: A loss of 12% gives SP 440. Find CP. $\rightarrow CP = 440 \times 100/88 = \mathbf{500}$.

6. Simple & Compound Interest

Core: SI = $PRT/100$. CI: $A = P(1+R/100)^T$. CI-SI (2 yrs) = $P(R/100)^2$.

Solved 1: SI on 6000 at 5% for 4 years. $\rightarrow 6000 \times 5 \times 4/100 = \mathbf{1200}$.

Solved 2: CI on 5000 at 10% for 2 years. $\rightarrow A = 5000(1.1)^2 = 6050 \Rightarrow CI = \mathbf{1050}$.

Solved 3: A sum doubles in 10 years at SI. Find rate. $\rightarrow SI = P \Rightarrow 100 = R \times 10 \Rightarrow R = \mathbf{10\%}$.

7. Time & Work / Work & Wages

Core: 1 day's work = $1/n$. Together = $AB/(A+B)$. Wages split in ratio of work done.

Solved 1: A in 10 days, B in 15 days; together? $\rightarrow AB/(A+B) = 150/25 = \mathbf{6 \text{ days}}$.

Solved 2: 8 men finish in 6 days. How long for 4 men? $\rightarrow \text{Work} = 8 \times 6 = 48 \text{ man-days}$; $48/4 = \mathbf{12 \text{ days}}$.

Solved 3: A and B finish a job for ₹600. A works $1/10$ per day, B $1/15$. Wages? $\rightarrow$ Efficiency ratio $1/10 : 1/15 = 3:2 \Rightarrow A = ₹360, B = ₹240$.

8. Time, Speed & Distance

Core: Speed = distance/time. $1 \text{ km/h} = 5/18 \text{ m/s}$. Train+pole: L/speed ; train+platform: $(L+P)/\text{speed}$. Relative speed: sum (opposite), difference (same direction). Boats: down = $b+s$, up = $b-s$.

Solved 1: 90 km/h to m/s? $\rightarrow 90 \times 5/18 = \mathbf{25 \text{ m/s}}$.

Solved 2: A 240 m train at 20 m/s crosses a 160 m bridge in? $\rightarrow (240+160)/20 = \mathbf{20 \text{ s}}$.

Solved 3: A boat goes 16 km downstream in 2 h; stream 2 km/h. Boat speed in still water? $\rightarrow$ Down speed = $8 \text{ km/h} = b+2 \Rightarrow b = \mathbf{6 \text{ km/h}}$.

9. Partnership

Core: Profit shared in ratio of (capital $\times$ time).

Solved: A invests ₹5000 for 12 months, B ₹6000 for 6 months. Profit ratio? → A: $5000 \times 12 = 60000$; B: $6000 \times 6 = 36000 \Rightarrow 60:36 = \mathbf{5:3}$.

10. Mensuration

Key formulas table:

Shape	Area	Perimeter / Volume
Rectangle	$l \times b$	$2(l+b)$
Square	a^2	$4a$
Triangle	$\frac{1}{2} \cdot b \cdot h$	—
Circle	πr^2	$2\pi r$
Cube	—	$V = a^3, SA = 6a^2$
Cuboid	—	$V = lbh$
Cylinder	—	$V = \pi r^2 h, CSA = 2\pi rh$
Sphere	—	$V = (4/3)\pi r^3, SA = 4\pi r^2$
Cone	—	$V = (1/3)\pi r^2 h$

Solved 1: Circle $r = 7$ ($\pi = 22/7$). Area? → $22/7 \times 49 = \mathbf{154}$.

Solved 2: Volume of a cylinder $r = 7, h = 10$. → $22/7 \times 49 \times 10 = \mathbf{1540 \text{ cubic units}}$.

11. Clocks & Calendars

Clock: Minute hand gains $5.5^\circ/\text{min}$ over hour hand. At 3:00 angle = 90° . Hands coincide every ≈ 65.45 min. **Calendar:** Ordinary year = 1 odd day; leap year = 2 odd days. Leap year: divisible by 4 (centuries by 400).

Solved: Angle between hands at 6:00? → $6 \times 30 = \mathbf{180^\circ}$.

Practice Strategy for Arithmetic

1. Master one topic's formula, then solve 20–30 questions before moving on.
2. Keep a single formula sheet; revise it daily.
3. Always check units (km/h vs m/s) and read whether the question wants profit on cost or on SP.
4. Time yourself — under 45 seconds per question is the target.

Ask for a similar deep-dive on Reasoning or any GS subject (e.g. full Indian Polity). Verify difficulty level against the official APSLPB notification.